11

Transfer matrix and diagonalisation of spin chains

11.1 Introduction

In Chapters 8 and 10 we studied methods for simulating classical systems consisting of many interacting degrees of freedom. In these methods, a sequence of system configurations is generated, and from this sequence averages of physical quantities, given as functions of the degrees of freedom (positions and momenta, or spins), can be determined. These quantities are called mechanical quantities, and the expressions of their expectation values are called *mechanical averages*.

There exist, however, quantities that cannot be determined straightforwardly using these methods. These quantities include free energies and chemical potentials. The point is that these quantities are not given as a *normalised* average, which for mechanical quantities is replaced by an average over the configurations generated in the simulation. In the previous chapters we have seen that it is not straightforward to find free energies and chemical potentials using MC and MD methods (see Section 10.5).

In this chapter we discuss a method which enables us to find free energies for lattice spin models with very high accuracy; this is the *transfer matrix method*. This method calculates the free energy of a model defined on a strip of finite width and infinite length directly in terms of the largest eigenvalue of a large matrix, the *transfer matrix*. This matrix contains essentially the Boltzmann weights for adding an extra row of spins to the strip. In Chapter 12 we shall see that the transfer matrix is the analogue of the time evolution operator in quantum mechanics. The fact that we can apply matrix methods to both quantum mechanics problems and statistical problems results from the fact that statistical and quantum mechanics are intimately related, as will be shown in the next chapter.

The transfer matrix is an operator acting in a dimension which is one lower than the dimension of the original, classical system. In this lower dimensional space, we must solve a quantum problem. This can be an interesting application by itself, even without reference to a transfer matrix relating the quantum problem to a classical one. We shall discuss one-dimensional quantum problems extensively in this chapter. Not too long ago, a new method was developed for treating this problem very efficiently. It is based on renormalisation and on the density matrix – hence its name, *density matrix renormalisation group*. We shall consider a few applications of this method in the last sections of this chapters.

In the next section we solve the one-dimensional Ising model exactly using the transfer matrix. We shall derive the free energy and the pair correlation function from this matrix. In the following section we shall describe the transfer matrix for two-dimensional spin models and see how transfer matrix techniques allow for numerically exact solutions of two-dimensional models on an infinite strip of finite width. This, in combination with finite-size scaling methods, makes the transfer matrix method very useful for obtaining accurate data for two-dimensional statistical models. For details concerning the numerical transfer matrix method, see Ref. [1].

Later sections are then dedicated to numerically diagonalising quantum chains (Section 11.5), to quantum renormalisation methods (11.6), and a particular form of this, the density matrix renormalisation group method (11.7).

11.2 The one-dimensional Ising model and the transfer matrix

In this section we present the exact solution of the one-dimensional Ising model. To arrive at the solution we must introduce a new concept, the *transfer matrix*, whose largest eigenvalue determines the partition function. The one-dimensional Ising model (see also Section 7.2.2) consists of a chain of spins numbered 1 through L; the spins assume values $s_i = \pm$, and the Hamiltonian is given by

$$\mathcal{H}[\{s_i\}] = \sum_{i=1}^{L}(-Js_i s_{i+1} - Hs_i). \tag{11.1}$$

J is the coupling strength between the spins, and the external field H favours spins of one particular sign. Periodic boundary conditions are imposed by identifying the spins at sites 1 and $L+1$. The partition function is

$$Z = \sum_{\{s_i\}} \exp\left[\beta \sum_{i=1}^{L}(Js_i s_{i+1} + Hs_i)\right]. \tag{11.2}$$

The sum runs over all possible spin configurations $\{s_i\}$.

The exponent can be written as a product over all nearest neighbour pairs:

$$Z = \sum_{\{s_i\}} \prod_{i=1}^{L} \exp[\beta J s_i s_{i+1} + \beta H s_i]. \tag{11.3}$$

It is therefore convenient to define the *transfer matrix* which contains the contribution of the pair s_i, s_{i+1} to the Boltzmann factor of the full chain:

$$T_{s_i,s_{i+1}} = \langle s_i|T|s_{i+1}\rangle = \exp[\beta(Js_is_{i+1} + Hs_i/2 + Hs_{i+1}/2)]. \tag{11.4}$$

The quantum mechanical Dirac notation is convenient here. The contribution of the magnetic field H has been symmetrically distributed over s_i and s_{i+1} in order to arrive at a symmetric transfer matrix.[1] The transfer matrix has the form:

$$T = \begin{pmatrix} e^{\beta(J+H)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} \end{pmatrix} \tag{11.5}$$

in a representation with basis vectors $|+\rangle = (1, 0)$ and $|-\rangle = (0, 1)$.

Now we can rewrite the partition function as

$$Z = \sum_{\{s_i\}} \langle s_1|T|s_2\rangle\langle s_2|T|s_3\rangle \cdots \langle s_L|T|s_1\rangle. \tag{11.6}$$

The sum is over all $s_i = \pm 1$, so we may write

$$Z = \sum_{s_1=\pm 1} \langle s_1|T^L|s_1\rangle = \mathrm{Tr}\,(T^L), \tag{11.7}$$

where we have used the completeness property $\sum_{s=\pm 1} |s\rangle\langle s| = I$; I is the 2×2 unit matrix.

The transfer matrix is a 2×2 matrix, which has two eigenvalues, λ_0 and λ_1. We then have

$$Z = \lambda_0^L + \lambda_1^L. \tag{11.8}$$

As the partition function is positive, the eigenvalue with the largest absolute value must be positive. Suppose that this eigenvalue is λ_0, then

$$Z = \lambda_0^L + \lambda_1^L \approx \lambda_0^L. \tag{11.9}$$

The approximation becomes exact for the $L \to \infty$, i.e. the thermodynamic limit. Therefore, using $F = k_B T \ln Z$, we obtain the following result for the free energy per spin:

$$F/L = -k_B T \ln \lambda_0. \tag{11.10}$$

In the case that both eigenvalues are equal, the result remains unaltered as can easily be checked. The transfer matrix method is trivially generalised to models with more than two possible values on each site along the chain – for example when the spins can assume more than two values, or when there is more than one spin per site. It is also possible to include further than nearest neighbour interactions. In these cases the matrix will be larger than 2×2, and numerical diagonalisation will be

[1] Other distributions of these interactions are also possible; the physical properties that we shall calculate are not affected by the choice we make here.

necessary, but for matrix sizes up to $10\,000 \times 10\,000$, diagonalisation of the transfer matrix is a trivial numerical exercise (Appendix A8.2).

Let us now calculate the expectation value of the spin at site m. As a result of translation invariance, the result is independent of m. This expectation value is given by

$$\langle s_m \rangle = \frac{\sum_{\{s_i\}} s_m \prod_{i=1}^{L} \exp[\beta J s_i s_{i+1} + \beta H s_i]}{\sum_{\{s_i\}} \prod_{i=1}^{L} \exp[\beta J s_i s_{i+1} + \beta H s_i]}. \tag{11.11}$$

Introducing again the transfer matrix and using the same argument as above to retain the largest eigenvalue λ_0 with eigenvalue ϕ_0 only, we obtain

$$\langle s_m \rangle = \frac{\sum_{s_1} \langle s_1 | T^{m-1} \hat{s}_m T^{L-m+1} | s_1 \rangle}{\sum_{s_1} \langle s_1 | T^L | s_1 \rangle} = \frac{\langle \phi_0 | \hat{s}_m | \phi_0 \rangle \lambda_0^L}{\lambda_0^L} = \langle \phi_0 | \hat{s} | \phi_0 \rangle. \tag{11.12}$$

In a spinor representation, in which the spin-up and -down states are $(1, 0)$ and $(0, 1)$ respectively, the operator $\hat{s}$ is the Pauli matrix σ_z. There is no subscript m in the rightmost expression as the ground state eigenvector ϕ_0 is independent of m.

We can also calculate the correlation function. This is defined as

$$g(m - n) = \langle s_n s_m \rangle - \langle s_m \rangle \langle s_n \rangle. \tag{11.13}$$

Denoting the eigenvalues of T by λ_α (and higher indices) and the corresponding eigenvectors by ϕ_α (for the one-dimensional Ising model, α assumes only the two values 0 and 1), we have

$$\begin{aligned} g(m-n) = \frac{1}{Z} \sum_{\phi_\alpha \cdots \phi_\epsilon} & \langle \phi_\alpha | T^{m-1} | \phi_\beta \rangle \langle \phi_\beta | \hat{s}_m | \phi_\gamma \rangle \langle \phi_\gamma | T^{m-n} | \phi_\delta \rangle \\ & \times \langle \phi_\delta | \hat{s}_n | \phi_\epsilon \rangle \langle \phi_\epsilon | T^{L-n+1} | \phi_\alpha \rangle - \langle \phi_0 | \hat{s} | \phi_0 \rangle^2. \end{aligned} \tag{11.14}$$

The last eigenvector ϕ_α in the first term is the same as the first eigenvector because of the periodic boundary conditions. One is usually interested in the long-range behaviour of the correlation function:

$$1 \ll |m - n| \ll L. \tag{11.15}$$

This suggests that the major contribution to the correlation function comes from replacing all eigenvalues in (11.14) by the largest one, λ_0. However, inspection shows that the two terms in (11.14) cancel exactly for this choice. The major contribution to the correlation function is therefore obtained by replacing the transfer matrices acting between positions n and m by the largest-but-one eigenvalue, λ_1. In the limit (11.15), all the other choices are much smaller. This results in

$$g(m - n) = |\langle \phi_0 | s | \phi_1 \rangle|^2 \left(\frac{\lambda_1}{\lambda_0} \right)^{|m-n|}. \tag{11.16}$$

We see that the correlation drops off exponentially, unless $\lambda_0 = \lambda_1$, that is, unless the largest eigenvalue is degenerate. This implies that critical behaviour can only occur when the largest eigenvalue is degenerate – recall that critical behaviour is characterised by power-law decay of correlation functions.

In this regard Frobenius' theorem is important. This states that the largest eigenvalue of a positive matrix (i.e. a matrix with all positive elements) of finite size is nondegenerate [2]; that is, our correlation functions will never show critical behaviour. The only way to obtain critical behaviour is by having a matrix of infinite size, which is only possible when the degrees of freedom on a site can assume an infinite number of different values. For a one-dimensional model on a lattice (chain) in which the degrees of freedom on a lattice site can assume a finite number of different values, the correlation function is always exponential.

In Chapter 12 we shall see that the analysis applied here carries over to quantum mechanics, the transfer matrix corresponding to the quantum mechanical time evolution operator $\exp(-\mathrm{i}tH/\hbar)$, where H is now the quantum Hamiltonian. If this Hamiltonian has eigenvalues E_α, we have $\lambda_\alpha = \exp(-\mathrm{i}tE_\alpha/\hbar)$. After an analytic continuation into imaginary time $t \to -\mathrm{i}t$, the eigenvalues become real: $\lambda_\alpha = \exp(-tE_\alpha/\hbar)$ and we see that degeneracy of the largest eigenvalue of the transfer matrix (or time-evolution operator) implies degeneracy of the quantum ground state energy. As an example, consider the one-dimensional Ising model in zero magnetic field. It is easy to see that for this model the quantum Hamiltonian occurring in the exponent of the time-evolution operator can be written as

$$H = -\tilde{J}\sigma_x, \tag{11.17}$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{11.18}$$

is the Pauli matrix, and $\tilde{J}$ is related to J by

$$\tanh(\beta\tilde{J}) = \exp(-2\beta J) \tag{11.19}$$

(see Problem 11.1).

In quantum field theory (see Chapter 15), the ground state is the vacuum (no particles present) and the first excited state is the state with one particle at rest. As the relativistic energy is given by $E = \sqrt{p^2c^2 + m^2c^4}$, the gap between these two states is given by the particle rest energy mc^2. Degeneracy of the ground state implies therefore that the particle mass is zero. A critical point is therefore often identified with a vanishing mass.

11.3 Two-dimensional spin models

We shall now describe the transfer matrix analysis for the two-dimensional Ising model; generalisation to models with more spin degrees of freedom is straightforward. Using the transfer matrix method we can calculate the free energy of an Ising model on an infinite strip of width M with periodic boundary conditions (PBC) connecting the two sides of the strip – it is therefore convenient to imagine the strip to be wrapped around a cylinder. The Hamiltonian of the Ising model in this geometry is given by

$$H = \sum_{i=1}^{M} \sum_{j=-\infty}^{\infty} (-J s_{i,j} s_{i,j+1} - J s_{i,j} s_{i+1,j} - H s_{i,j}), \tag{11.20}$$

where $s_{1,j} \equiv s_{M+1,j}$ as a result of the PBC.

For this model, the transfer matrix is the contribution to the Boltzmann factor of two adjacent lattice rows (the transfer matrix acts along the axis of the cylinder; the rows are perpendicular to this direction). The possible states of the rows are the indices of the transfer matrix. If the rows contain M sites, the size of the transfer matrix is $2^M \times 2^M$. We represent the configuration of row j by the state $|S_j\rangle$:

$$|S_j\rangle = |s_{1,j} s_{2,j} \dots s_{M,j}\rangle = |s_{1,j}\rangle \otimes |s_{2,j}\rangle \otimes \dots \otimes |s_{M,j}\rangle. \tag{11.21}$$

The transfer matrix for rows j and $j+1$ is found as

$$\begin{aligned}\langle S_j|T|S_{j+1}\rangle = \exp\Bigg[&\beta \sum_{i=1}^{M} J \left(\frac{1}{2} s_{i,j} s_{i+1,j} + \frac{1}{2} s_{i,j+1} s_{i+1,j+1} + s_{i,j} s_{i,j+1} \right) \\ &+ \frac{\beta}{2} H \sum_{i=1}^{M} (s_{i,j} + s_{i,j+1}) \Bigg],\end{aligned} \tag{11.22}$$

The eigenvalues can now be found from a numerical diagonalisation of this matrix: its largest eigenvalues (in absolute value) determine the free energy and correlation functions of the model on a semi-infinite strip of width M.

Diagonalisation can be performed straightforwardly up to matrices of size $10\,000 \times 10\,000$ (and beyond if one uses powerful machines). However, we need only the largest few eigenvalues and there exist special numerical methods for calculating these which, for sparse matrices, are much more efficient than standard methods. Most convenient is Lanczos' method, which is described in Appendix A8.2. In this method, the matrix enters only via the multiplication with a vector, and this multiplication can be carried out efficiently, in particular for a sparse matrix, provided we take only the nonzero entries into account. Unfortunately, the transfer matrix of the Ising model is not sparse at all – it follows from Eq. (11.22) that it has a nonzero value for each pair of possible row configurations S_j and S_{j+1}.

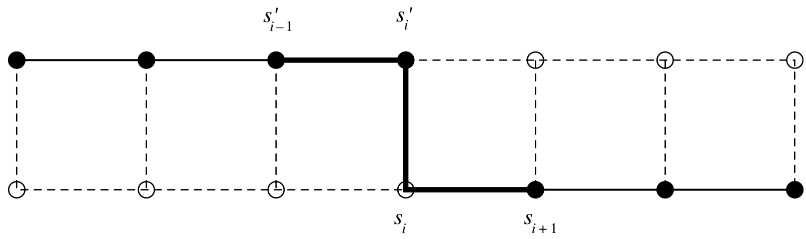

Figure 11.1. A step in the multiplication of a vector with the transfer matrix for the Ising case.

There is, however, a way to perform matrix–vector multiplications efficiently, using the fact that T factorises into a product of sparse matrices:

$$T = T_M T_{M-1} \cdots T_1 \tag{11.23}$$

where the matrix T_i is associated with site number i. To understand the explicit form of the submatrices T_i, it is useful to look at Figure 11.1. Let us call ϕ the state with matrix elements corresponding to the bottom row in that figure. Calling the bottom row configurations $|S\rangle$, and the top row $|S'\rangle$, we want to calculate the elements of the vector $|\psi\rangle = T|\phi\rangle$:

$$\langle S'|\psi\rangle = \langle S'|T|\phi\rangle = \sum_S \langle S'|T|S\rangle\langle S|\phi\rangle \tag{11.24}$$

for each top row configuration S'. The full matrix T contains couplings within the two horizontal rows and on the vertical links between them. The submatrices T_i contain only the couplings on the thick lines in Figure 11.1. We could therefore say that the application of a matrix T_i replaces the lower spin s_i by the upper spin s'_i, leaving the remaining spins unchanged.

We can now give the form of the matrices T_i. It acts between two states which are somehow intermediate between S and S': the state Σ' on the left of T_i represents the spins $s'_1, s'_2, \ldots, s'_i, s_{i+1}, \ldots, s_M$, and the state Σ on the right represents the spins $s'_1, s'_2, \ldots, s'_{i-1}, s_i, \ldots, s_M$. In terms of these states, which have elements σ_j and σ'_j respectively, we have

$$\begin{aligned}
\langle \Sigma'|T_i|\Sigma\rangle &= \langle \sigma'_1 \ldots \sigma'_{i-1}\sigma'_i\sigma'_{i+1} \ldots \sigma'_M|T_i|\sigma_1 \ldots \sigma_{i-1}\sigma_i\sigma_{i+1} \ldots \sigma_M\rangle \\
&= \exp\left\{\beta J\left[\frac{1}{2}(\sigma_i\sigma_{i+1} + \sigma'_i\sigma'_{i-1}) + \sigma_i\sigma'_i\right]\right\} \\
&\quad \times \delta_{\sigma_1\sigma'_1}\delta_{\sigma_2\sigma'_2} \ldots \delta_{\sigma_{i-1}\sigma'_{i-1}}\delta_{\sigma_{i+1}\sigma'_{i+1}} \ldots \delta_{\sigma_M\sigma'_M}.
\end{aligned} \tag{11.25}$$

The horizontal couplings have a factor $1/2$ because they are taken into account twice: once when the transfer matrix couples the previous row to the current row, and once when the transfer matrix couples the present row to the next one. The form given in (11.25) is not correct for $i = 1$ or $i = M$, because the first new spin, s'_1, does not yet have a left neighbour. Therefore, in the matrix T_1, the term $\beta J\sigma'_1\sigma'_M/2$ is left out and replaced by $\beta J\sigma_1\sigma_M/2$. Similarly, for $i = M$, the coupling $\beta J\sigma_M\sigma_1/2$ is replaced by $\beta J\sigma'_M\sigma'_1/2$.

How do we perform the full matrix–vector multiplication? We introduce a set of intermediate vectors $\psi^{(i)}$, defined by

$$|\psi^{(0)}\rangle \equiv |\phi\rangle \tag{11.26a}$$

$$|\psi^{(i)}\rangle \equiv T_i|\psi^{(i-1)}\rangle, \tag{11.26b}$$

so that $|\psi^{(M)}\rangle = |\psi\rangle$. Suppose we have the vector $|\psi^{(i-1)}\rangle$ at our disposal, that is, we have all its matrix elements $\langle S^{(i-1)}|\psi^{(i-1)}\rangle$ stored in an array (below we shall describe how the different configurations S correspond to the array index). The multiplication of the vector $|\psi^{(i-1)}\rangle$ by the matrix T_i is done as follows ($\Sigma \equiv S^{(i-1)}$; $\Sigma' \equiv S^{(i)}$):

FOR all states $S^{(i)}$ DO
  FOR all $S^{(i-1)}$ equal to $S^{(i)}$ except for spin number i DO
    Calculate $\langle S^{(i)}|\psi^{(i)}\rangle = \langle S^{(i)}|T_i|S^{(i-1)}\rangle\langle S^{(i-1)}|\psi^{(i-1)}\rangle$.
  END FOR
END FOR

To do the evaluation in the inner loop, we need the vector elements $\langle S^{(i-1)}|\psi^{(i-1)}\rangle$ and the matrix elements $\langle S^{(i)}|T_i|S^{(i-1)}\rangle$. The former are stored in an array, and the latter are given by (11.25).

It now remains to find a relation between array and loop indices, and the states S. This is done using binary encoding, using 'bits' b_i which assume values 0 or 1:

$$n = b_0 + b_1 2^1 + b_2 2^2 + \cdots + b_{M-1}2^{M-1} \tag{11.27}$$

where $b_i = 1$ corresponds to $s_{i+1} = 1$ and $b_i = 0$ to $s_{i+1} = -1$. The matrix element $\langle S^{(i)}|T_i|S^{(i-1)}\rangle$ depends on $s_{i-1}(= s'_{i-1})$, $s_{i+1}(= s'_{i+1})$, and on s_i and s'_i (the latter are not necessarily equal). For an integer n, the ith bit can be found as $b_{i-1} = (n/2^{i-1}) \mod 2$, where integer division is assumed. Most computer languages have, however, an intrinsic function returning the value of any desired bit of an integer number. It should be noted that in the algorithm given above, the central evaluation for the first and last sites (1 and M) differs slightly from the other ones because the respective transfer matrix elements were defined differently (see above).

Having a routine for evaluating the transfer matrix–vector product, this can be used in the Lanczos algorithm in order to find the lowest eigenvalue and vector.

PROGRAMMING EXERCISE

Write a program for calculating the largest eigenvalue of the transfer matrix of the Ising model. The program involves a considerable amount of book-keeping and the correctness can be tested only in the very end, when the free energies are actually calculated. When debugging, it is advisable to reserve just sufficient memory for the rows and then compile the code with the range-check option on (in this case it is checked whether indices of arrays addressed in the program lie within the appropriate bounds). Furthermore, an error in the multiplication routine will usually result in the transfer matrix not being Hermitian, which in turn causes the Lanczos routine to converge very slowly, if at all. For strip widths M somewhere between 4 and 12, the Lanczos routine should converge within less than 20 steps.

Check For $M = 6$ and the critical coupling $J/(k_B T) = 0.440\,687$, you should find $\lambda_0 = 276.6004$.

If the program works correctly, we can calculate the free energy. First of all, rather than considering the free energy $F = \ln \lambda_0$ (we leave out the factor $k_B T$), we calculate the free energy *per spin*, given as $f = (\ln \lambda_0)/M$, which should converge to a fixed value for large widths. To analyse the results using finite size scaling, we quote a few results from conformal field theory [3, 4]. In this theory, two-dimensional critical models are labelled by a number, the *central charge*, c, and the possible values of the central charge smaller than one form a discrete set.[2]

The central charge also has a physical interpretation. If we consider the model on a strip of width M, the free energy will scale approximately proportional with M. However, in diagonalising the transfer matrix (or the Hamiltonian), changing M means changing the boundary conditions, and therefore there will be corrections to this scaling behaviour. This mechanism occurs also in electrostatics [5], where it is known as the Casimir effect: two parallel conducting plates will attract each other: the energy per area scales as Const $\times M^{-3}$ (zero temperature). It turns out that in our case, the statistical model on a strip, the free energy (per unit length along the strip) scales as

$$F = \ln \lambda_0 = f_{\text{Bulk}} M - \frac{\pi c}{6} \frac{1}{M}. \tag{11.28}$$

f_{Bulk} is the free energy per site of the infinite system. For the Ising model it is known that the central charge is equal to $1/2$. Each value of c defines a set of possible critical exponents of the theory, and indeed, the thermal and magnetic exponent of the Ising model, with values 1 and 1/8 respectively, are in the set for $c = 1/2$. The determination of the central charge is therefore very useful to classify the model's critical behaviour.

[2] For $c > 1$ there are arguments from supersymmetric field theory that also suggest discrete series, but they are not based on requirements that should hold rigorously.

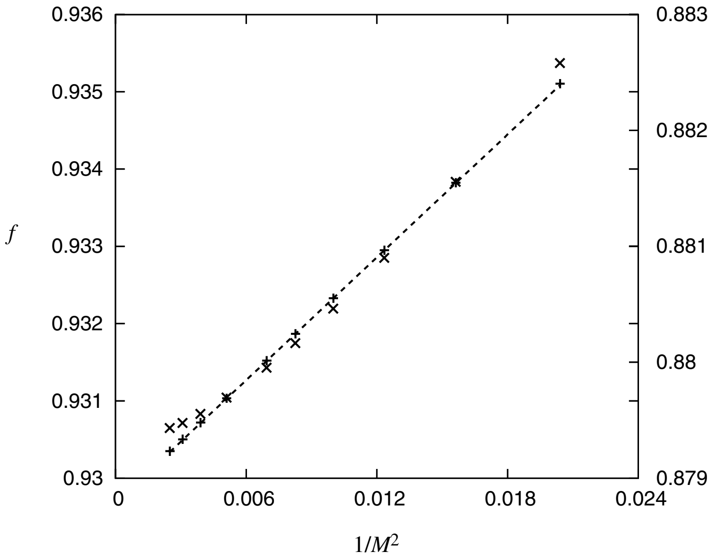

Figure 11.2. Free energy per site of the Ising model on a strip as a function of $1/M^2$. Values for M run from 7 to 20. The pluses correspond to the critical point $J = 0.4407$; the crosses (right axis) are the data for an off-critical point: $J = 0.4$. The straight line through the data has slope $\pi/12$.

Therefore, if we plot our results for the Ising model in the form $(\ln \lambda_0)/M$ vs. $1/M^2$ we should obtain a straight line with slope $\pi/12$. Figure 11.2 shows that this is indeed the case. As this is a finite size correction to the bulk free energy, the eigenvectors λ_0 must be determined with a precision of about six significant digits.

The magnetic exponent can be obtained by comparing the free energies for systems with periodic and antiperiodic boundary conditions, see Ref. [4]. Antiperiodic means the bonds across a seam along the cylinder are antiferromagnetic. The free energy difference then scales with the width M as

$$F_{\mathrm{AP}} - F_{\mathrm{P}} = \frac{2\pi}{M} x, \tag{11.29}$$

where x is the magnetic exponent – indeed, $x = 1/8$ can be found in this way for the Ising model.

11.4 More complicated models

The linear size of the transfer matrix increases exponentially with the strip width. This puts severe limits on the system sizes that can be treated with this method. In particular, this prohibits calculations with reasonable system size for models with larger numbers of degrees of freedom. There exist models, such as the clock or the

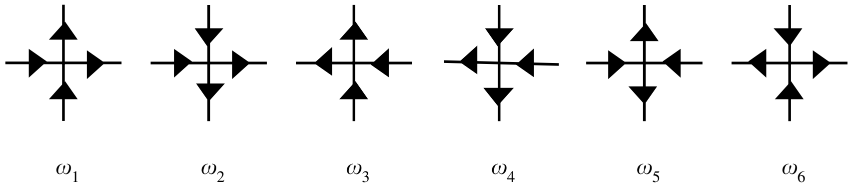

Figure 11.3. Vertex configurations of the six-vertex model. The Boltzmann weight of a lattice configuration is the product of the weights ω_i for all sites.

Potts model, with q spin states per site, where q can be any number,[3] and the transfer matrix method is only feasible for q-values up to about 4. It turns out that many of these models can be mapped exactly onto models with fewer degrees of freedom, and the transfer matrix method can then be applied to the latter in order to obtain critical exponents. This approach was used by Blöte and Nightingale [1, 3, 8] to analyse the Potts model. They used a mapping [7, 9, 10] of the Potts model onto the six-vertex model [11].

In the six-vertex model, the degrees of freedom do not reside on the sites, but on the links of a square lattice. They are two-valued and are represented by arrows on the links. For each site, there are in principle $2^4 = 16$ different configurations of arrows on the four links connected to that site. This number is, however, reduced if we require that the net flow into the sites is zero: the number of incoming and outgoing arrows must be equal for each site. This leaves only six different site configurations or vertices, hence the name of the model. The different vertices are shown in Figure 11.3. Each vertex is assigned a weight, and the Boltzmann weight of a lattice is the product of the vertex weights of all sites. Often the case $\omega_1 = \omega_2 = \omega_3 = \omega_4$; $\omega_5 = \omega_6$ is considered. The model is called the 'F-model' and this particular case has been solved exactly by Lieb [11]. The F-model turns out to be critical, with an infinite order transition (of the Kosterlitz–Thouless type, see Section 15.5.2) for $2\omega_1 = \omega_5$. In Problem 11.2, a transfer matrix implementation for the standard six-vertex is described. In the case of the Potts model, the equivalent six-vertex model is a slightly modified version of the original model, and this has as a consequence that the transfer matrix is no longer Hermitian for q noninteger, which complicates its diagonalisation (for details, see Ref. [1]).

If a transformation onto a simple model is not known, application of the transfer matrix for models in which the degrees of freedom can assume many values might still be possible, using a technique inspired by quantum Monte Carlo methods [12]. This method will be considered in some detail in Section 12.6.

[3] Even noninteger values of q are relevant [6, 7].

11.5 'Exact' diagonalisation of quantum chains

So far in this chapter, we have studied methods for diagonalising the transfer matrices of two-dimensional models. This is equivalent to diagonalising Hamiltonians of one-dimensional quantum systems. In the next few sections we shall consider the analysis of one-dimensional quantum chains more systematically.

You may be sceptical about the use of studying one-dimensional quantum chains. There are, however, good reasons for considering such systems. First, they are equivalent to classical systems in two dimensions, as we have seen extensively in the first half of this chapter. Second, there exist experimental realisations of quasi-one dimensional systems in some particular crystals; see for example Ref. [13]. Last but not least, the fact that one-dimensional chains can be studied successfully using analytical and computational tools makes them useful as testing grounds for new methods which may be successful in higher dimensions too. A nice introduction to the material covered in this and the following sections is Ref. [14].

The quantum chains which are studied most intensively are spin chains and Hubbard-like systems. Spin chains consist of quantum spins located on a one-dimensional lattice. The magnetic spin quantum numbers can assume values $-S$, $-S+1$ up to S, where the maximal spin S is a positive, (half-)integer number. Note that from now on we shall simply call 'spin' the eigenvalue of the z-component of the spin-angular momentum operator (without the factor $\hbar$). A famous example is the Heisenberg (anti-)ferromagnet, which is described by the Hamiltonian:

$$H = J \sum_{i=1}^{L-1} \mathbf{S}_i \mathbf{S}_{i+1}. \tag{11.30}$$

This is a chain with *open ends* as the first and last spins couple only to a single neighbour. If we connect the first and last spin by putting $\mathbf{S}_{L+1} \equiv \mathbf{S}_1$, we have a periodic chain. For positive values of J the chain is antiferromagnetic.

The one-dimensional Hubbard model describes fermions[4] on a chain. The fermions are spin-1/2 particles and because of the Pauli principle there are on each site four possibilities: zero particles, one spin-up or one spin-down particle, or both. The motion of the particles along the chain is represented by a hopping term. There is also a Coulomb interaction, which is assumed to be strongly local: only particles on the same site can feel it. For particles on different sites, the Coulomb interaction is neglected. The Hamiltonian is

$$H = -t \sum_{\sigma;i=1}^{L-1} (c^{\dagger}_{i,\sigma} c_{i+1,\sigma} + c^{\dagger}_{i+1,\sigma} c_{i,\sigma}) + U \sum_{i=1}^{L} n_{i,\uparrow} n_{i,\downarrow}. \tag{11.31}$$

[4] Bose–Hubbard models are presently also the subject of research, but the fermion version has been studied most intensively.

The real parameter t represents the hopping rate, and U (which is also real) describes the Coulomb interaction. The fermion creation and annihilation operators $c_{i,\sigma}$ and $c_{i,\sigma}^{\dagger}$ have a site index i and a spin index $\sigma = \pm 1/2$ (these two values are denoted as $\uparrow$ and $\downarrow$ in the Coulomb term). They satisfy the usual fermion anticommutation relations:

$$\{c_{i,\sigma}^{\dagger}, c_{j,\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}. \tag{11.32}$$

The other anticommutators vanish. Finally, the number-operators $n_{i,\sigma}$ are given by

$$n_{i,\sigma} = c_{i,\sigma}^{\dagger} c_{i,\sigma}. \tag{11.33}$$

Other models, such as the $t - J$ model, can be related to the Hubbard model, and this model reduces in a particular limit to the Heisenberg chain; we shall not go into details but refer to the literature [15].

In the following, we shall mainly restrict ourselves to the Heisenberg chain, which serves to illustrate the numerical methods suitable for studying quantum chains; we shall briefly mention how these can be adapted to Hubbard-like models where appropriate.

11.5.1 Lanczos diagonalisation of the Heisenberg model

It is quite straightforward to diagonalise the Hamiltonian for the Heisenberg model – this is done using the Lanczos method (see Appendix A8.2). The main problem is writing a procedure for multiplying the Hamiltonian by some given vector $|\psi\rangle$.

The basis vectors of an L-site chain can be chosen as

$$|\Sigma\rangle = |s_1, s_2, \ldots, s_L\rangle, \tag{11.34}$$

where the s_i are the spins. For a spin-1/2 chain these spins assume the values $\pm 1/2$, and for the $S = 1$ chain 1, 0 and -1. The dimension of the Hilbert space is $(2S+1)^L$. We shall not use this spin representation in our program, but use a mapping of each basis state to an integer just as in the transfer matrix program. First, we change notation and let s_i run from 0 to $2S$ instead of from $-S$ to S. We write

$$|K\rangle = |s_1, s_2, \ldots, s_L\rangle, \tag{11.35}$$

where the integer K is given as

$$K = \sum_{i=1}^{L} s_i M_S^{i-1}; \tag{11.36}$$

$M_S = 2S + 1$. It is instructive to calculate the numbers K for all possible states of a $L = 4$ spin-1/2 chain, for which $s_i = 0, 1$, and the reader is invited to do this.

Before we proceed, we note that the Heisenberg Hamiltonian can be written in the form (see Problem 11.3)

$$H = J \sum_{i=1}^{N-1} \left[\frac{1}{2}(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+) + S_i^z S_{i+1}^z \right]. \tag{11.37}$$

Furthermore, the Hamiltonian commutes with quite a few symmetry operators. We shall study these in detail in Section 11.5.2.

Multiplication of this Hamiltonian with a given vector $|\psi\rangle$ proceeds in a way analogous to the transfer matrix problem: we 'zip' through the chain and collect all the generated terms into the resulting vector. The heart of the algorithm looks as follows:

```
FOR each site I DO
    FOR N1 = 0 TO (M_S)^(I-1) - 1
        FOR N2 = 0 TO (M_S)^(L-I-1) - 1
            FOR S1 = 0, M_S - 1
                FOR S2 = 0, M_S - 1
                    FOR S1P = 0, M_S - 1
                        FOR S2P = 0, M_S - 1
                            J = N1+N2*(M_S)^(I+1) + S1*(M_S)^(I-1) +
                                                S2*(M_S)^I;
                            JP = N1+N2*(M_S)^(I+1) + S1P*(M_S)^(I-1) +
                                                S2P*(M_S)^I;
                            HPsi(JP) = HPsi(JP)+
                                                HMat(S1P, S2P, S1, S2)*Psi(J);
                        END FOR;
                    END FOR;
                END FOR;
            END FOR;
        END FOR;
    END FOR;
END FOR;
```

It is understood that for i near the boundary, some modification is necessary. The matrix HMat is the part of the Hamiltonian which couples the spins at sites i and $i + 1$. This matrix is directly found from the Heisenberg Hamiltonian with $\mathbf{S}_i = \frac{1}{2}(\sigma_x, \sigma_y, \sigma_z)$, and σ_i are the three Pauli matrices.

This procedure can now readily be used in the Lanczos algorithm to find the lowest few eigenvalues.

PROGRAMMING EXERCISE

Program a routine for multiplying the Hamiltonian by a vector. Use this in a Lanczos algorithm to find the ground state and first excited state of a spin-1/2 antiferromagnetic Heisenberg chain. The infinite chain has no energy gap [16]; therefore, the energy gap should decrease with increasing length.

Check For a chain with 12 sites, you should find -5.3873909 for the ground state energy of a periodic chain.

The program can easily be extended to the spin-1 chain. You can then check the value of the Haldane gap [17] between the ground state and the first excited state, which is $0.4107 + 67.9/L^2$ *per site* for an long chain [18]. Note that to extend this program to the Hubbard model, M_S must be four (as there are four possible states for each site) and the Hamiltonian must be adjusted to that model.

11.5.2 Exploiting symmetry

Up to this point, we have hardly mentioned the use of symmetry when diagonalising a Hamiltonian. This is, however, extremely important, as the use of symmetry can result in a huge reduction of the CPU time needed for a calculation. To see this let us assume that we have some symmetry operation, represented by a quantum operator O which commutes with the Hamiltonian. In that case we can organise the eigenstates of the Hamiltonian such that they are also eigenstates of O and vice versa. Now suppose that we can easily find a complete set of eigenstates of O (this often turns out be the case). These states are then organised into subspaces, one for each eigenvalue of O. States within such a (possibly degenerate, and therefore multidimensional) eigenspace will be invariant under action of the Hamiltonian on them (see Problem 11.3). This means that the Hamiltonian will acquire a *block-diagonal* form when formulated with respect to this complete set. This is an extremely useful result. If we have for example a basis on a one-dimensional axis, and a Hamiltonian which is invariant under reflection, we divide up our N basis vectors into a subset of even and one consisting of odd basis functions. If there are equal numbers of each type, the Hamiltonian will have two blocks of size $N/2$ on the diagonal.

What is the use of this? Matrix diagonalisation is an order-N^3 algorithm. Diagonalising two $N/2 \times N/2$ matrices is therefore four times as fast as diagonalising a single $N \times N$ matrix. The invariant subspaces in which we solve our small Hamiltonian matrices are called *sectors*. Studying symmetries and their use in physics is mainly the domain of group theory. The interested reader is advised to consult specialised literature [19]. Here we shall implement the symmetries using pedestrian methods.

For the one-dimensional Heisenberg chain, the implementation of symmetry is very instructive. The symmetry operations are easy to find [20]. The symmetry operations we shall consider are translation symmetry (which holds for the periodic chain), conservation of angular momentum along the z axis (which results from rotational symmetry around that axis), and reflection symmetry (left-right symmetry). In Problem 11.3, these symmetries are considered in some detail.

We can find so many small sectors that it is possible to find *all* eigenvalues (thus not only the lowest ones) for the $L = 12$ Heisenberg chain of the previous section very quickly. We start by considering the rotation symmetry which leads to the conservation of the total angular momentum along the z-axis. This is

$$S_z^{\text{tot}} = \sum_{i=1}^{L} s_i. \tag{11.38}$$

It is clear that the states $|s_1, \ldots, s_L\rangle$ are eigenstates of the S_z^{tot}. We can simply loop over all the states $|K\rangle$ (K given by (11.36)), determine their total spin component along the z axis and reshuffle them into sets of equal S_z. You can try this for the Heisenberg chain and see that it leads to much smaller matrices. These matrices can already be diagonalised on a standard PC. However, an additional major improvement is realised by implementing translation symmetry (which only holds for the periodic case). Eigenstates of the translation operator, which shifts all spins one site to the right, are Bloch states. In fact, for each basis state $|\psi\rangle$ we can generate a Bloch state which is an eigenvector of the translation operator T as follows:

$$|\psi_k\rangle = |\psi\rangle + \mathrm{e}^{2\pi \mathrm{i}k/L} T|\psi\rangle + \mathrm{e}^{4\pi \mathrm{i}k/L} T^2|\psi\rangle + \cdots + \mathrm{e}^{2\pi \mathrm{i}(L-1)k/L} T^{L-1}|\psi\rangle. \tag{11.39}$$

It can easily be checked that the eigenvalue is $\mathrm{e}^{-2\pi \mathrm{i}k/L}$.

Let us now work out these symmetries for a chain consisting of four spins. The basis states can be grouped into states related to each other by translations.

Sz	0	1	2		3	4
k=0	$\|0\rangle$	$\|1\rangle+\|2\rangle+\|4\rangle+\|8\rangle$	$\|3\rangle+\|6\rangle+\|12\rangle+\|9\rangle$	$\|5\rangle+\|10\rangle$	$\|7\rangle+\|14\rangle+\|13\rangle+\|11\rangle$	$\|15\rangle$
k=1		$\|1\rangle+i\|2\rangle-\|4\rangle-i\|8\rangle$	$\|3\rangle+i\|6\rangle-\|12\rangle-i\|9\rangle$		$\|7\rangle+i\|14\rangle-\|13\rangle-i\|11\rangle$	
k=2		$\|1\rangle-\|2\rangle+\|4\rangle-\|8\rangle$	$\|3\rangle-\|6\rangle+\|12\rangle-\|9\rangle$	$\|5\rangle-\|10\rangle$	$\|7\rangle-\|14\rangle+\|13\rangle-\|11\rangle$	
k=3		$\|1\rangle-i\|2\rangle-\|4\rangle+i\|8\rangle$	$\|3\rangle-i\|6\rangle-\|12\rangle+i\|9\rangle$		$\|7\rangle-i\|14\rangle-\|13\rangle+i\|11\rangle$	

All the boxes represent sectors of the Hilbert space: we see that only two sectors are two-dimensional; the others contain only a single state. This means that the diagonalisation can now easily be done by hand.

Note that some boxes are empty: this is because constructing the eigenstate according to the above recipe leads to the zero state. This happens when translating the state over one or two positions turns the state into itself. As the period of these states is smaller than four, the corresponding k-values are spaced by more than one. Note that the prefactors in the different states are given by $\exp(2\pi \mathrm{i}kl/L)$, where k is the k-vector, and this prefactor is for the state obtained by translating the first basis state over l sites. It is clear that shifting k by L leaves the states invariant.

We now consider reflection symmetry. This leads to combinations of states with wave vectors k and $-k$. Taking the periodicity $k \to k + L$ into account, we see that the $k = 3$ line above can be eliminated (it is equivalent to $k = 1$) and that we can take either even or odd combinations of $k, -k$ pairs. In the first case, the prefactor $\exp(2\pi \mathrm{i}kl/L)$ is replaced by $\cos(2\pi kl/L)$ and in the second case by $\sin(2\pi kl/L)$.

The program is now set up as follows. A loop over all states is performed. For each new state, the states which can be obtained by translation over 1, 2 and 3 (more generally 1 to $L - 1$) positions are considered; if such a state is identical to the first state, we know the periodicity of the state, and hence the nonzero k-vectors. Consider the $|0\rangle$ state as an example. A translation over one site transforms the state into itself – hence, the period in the k-values is $L/1 = L$. This is why we find only one state in the second and rightmost columns of the table above. Similarly, the state $|5\rangle$ has period two, hence we find only two nonzero states in the fifth column. All the other states have period 4, hence period 1 in the acceptable values of k.

Also, the spin S_z^{tot} of each state is determined (note that the translations do not affect this value). We now label the sectors by S_z, k (we omit the superscript 'tot' with S_z). Each time we find a state in this sector we add it to the basis of that sector. Note that we consider only $k = 0, \ldots, L/2$, and construct an even and an odd set (using cosine and sine as indicated above) for each sector S_z, k. When we have gone through all states in this way, we have complete basis sets for each sector. In a following step, we construct the Hamiltonian block matrix $\langle K|H|K'\rangle$ for each sector, where $|K\rangle$ and $|K'\rangle$ are basis states for the sector under consideration. For this calculation we use the procedure written in the previous subsection for calculating $H|K'\rangle$. Then we feed the resulting matrices into a library routine for diagonalising symmetric matrices.

PROGRAMMING EXERCISE

Write a program which implements these symmetries and calculates the full spectrum of the antiferromagnetic spin-1/2 Heisenberg chain.

Check You can check you results by comparing the lowest states with those obtained in the previous section. Furthermore, a useful check is whether the dimensions of all sectors add up to $(M_S)^L$. Also, you can run the program for $L = 4$ and check whether the sectors and basis vectors correspond to those given above, in the Table.

You have now seen group theory 'at work'. The possible improvements in performance are really impressive – for this reason, group theoretical methods are very important in computational physics. A particular example is solid state physics; in Chapter 6 we have already briefly mentioned the use of symmetry, in particular to reduce the Brillouin zone to the so-called 'irreducible wedge'. A nice discussion of the symmetry-issues in quantum systems can be found in Ref. [21].

11.6 Quantum renormalisation in real space

Calculations for systems whose behaviour is characterised by large length scales are usually time-consuming. Often, the systems have short-range interactions, but these generate correlations over large distances. The most obvious example is the Poisson equation for a point charge:

$$\nabla^2 V(\mathbf{r}) = \delta(\mathbf{r}). \tag{11.40}$$

Discretising the Laplace operator on a grid in a first order approximation couples only nearest neighbour lattice sites. On the other hand, the solution has long range character. In Appendix A7.2 we consider the multigrid method for treating this problem. This method uses successive coarsenings and refinements in order to find the solution very efficiently.

The multigrid method is reminiscent of the renormalisation method for critical phenomena. There we perform successive coarse grainings of the system, in order to extract its long-wavelength behaviour which is responsible for the occurrence of critical phenomena. In renormalisation procedures, we usually tend to throw away details relating to the short-wavelength behaviour. This causes the critical point to be found at the wrong value, but critical exponents are still found correctly or to a good approximation.

Often we are interested in the full solution, and not only the long-range behaviour. In these cases we must still treat the short-range behaviour accurately. The multigrid method accomplishes this by the refinements which alternate with the coarsenings. Another possible approach, which is closer to the standard renormalisation procedure, is to take a small sample system and extend the size of this and see whether it is possible to find a self-consistent limit which gives us the solution of the infinite system. Wilson has followed this approach with great success for the Kondo problem [22]. We shall not treat the Kondo problem in detail, but emphasise at the same time that the reason why this numerical renormalisation procedure worked so well is due to the special structure of the Kondo Hamiltonian, which contains couplings that decay exponentially with distance.

The lack of success when applying the renormalisation procedure to other problems has been nagging researchers until White and Noack [18, 23] came up with

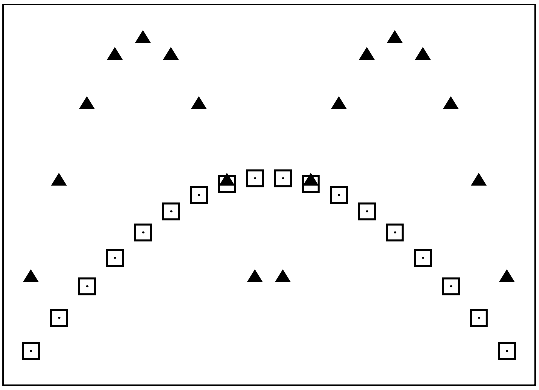

Figure 11.4. Two fixed-boundary solutions for a narrow box (solid triangles) and the ground state (open squares) for a large box. It is clear that the approximation of the ground state will be poor.

a solution for a quantum particle in a potential and for a quantum spin chain. The particle in a potential illustrates where the standard renormalisation procedure fails [18]. Consider a particle in a box with impenetrable walls (infinite potential outside the box). An idea for solving this problem efficiently is first to solve the problem in a box of half the width of the full box and then solve the problem for the full box using the ground state wave functions for the half-width boxes as basis functions in a variational calculation. Figure 11.4 illustrates why this method fails. The Schrödinger equation is solved for fixed boundary conditions, and building the ground state of the full problem from two functions that vanish at the centre is not efficient.

In order to improve the method, White and Noack built the solution of the larger box from solutions obtained for a smaller box *with different kinds of boundary conditions*. Consider two solutions, one of which has fixed zero boundary conditions on both sides, whereas the other has an open boundary condition on one side. The first has value zero, but (in general) a nonzero derivative. The second one has nonzero value at the open boundary, but its derivative is zero. *Any* type of boundary condition can be obtained by linear combination of the two solutions. This notion leads to the following algorithm.

1. Find the solutions of a small box, with fixed and open boundary conditions on both sides (four cases in total). Keep the lowest $M/4$ eigenstates of the Hamiltonian for each case.
2. Construct a Hamiltonian for the large box (twice as large as the small box) with respect to the basis found in step 1, for the same four cases as in step 1. See below for how this is done. The dimension of these Hamiltonians is $2M$.

3. Calculate the eigenstates of these four matrices and keep the lowest M. Now the large box is considered as a small box, and we proceed with step 1.

We shall now fill in the details of the algorithm. When having the small matrices at our disposal, we must construct the large matrices from these. For a large matrix with boundary conditions b and b′ (the b's denote 'fixed' or 'free') on the left and right hand side respectively, the recipe is

$$H^{\mathrm{L}}_{\mathrm{bb}'}(2M) = \begin{pmatrix} H^{\mathrm{S}}_{\mathrm{b,fixed}}(M) & T^{\mathrm{S}}(M) \\ (T^{\mathrm{S}})^{\dagger}(M) & H^{\mathrm{S}}_{\mathrm{fixed,b}'}(M) \end{pmatrix}, \tag{11.41}$$

where we have used the superscripts 'S' and 'L' to denote the matrices for the short and the long chain respectively. The arguments (M) and $(2M)$ specify the sizes of the matrices. We must carry out a similar procedure for the off-diagonal matrix:

$$T^{\mathrm{L}}(2M) = \begin{pmatrix} 0 & 0 \\ T^{\mathrm{S}}(M) & 0 \end{pmatrix}. \tag{11.42}$$

For each of the four possibilities $(\mathrm{b}, \mathrm{b}')$ we diagonalise the large matrix, yielding $2M$ eigenvectors for each case. We keep only the $M/4$ corresponding to the lowest energies, yielding a set of M basis states. These states are not necessarily orthogonal, so we use an orthogonalisation procedure such as Gram–Schmidt to transform them into an orthonormal basis. The matrices of the large box are now reduced to size $M \times M$ according to the standard procedure:

$$H^{\mathrm{L}}_{\mathrm{bb}'}(M) = V^{\dagger} H^{\mathrm{L}}_{\mathrm{bb}'}(2M) V, \tag{11.43}$$

where V is an $2M \times M$ matrix whose M columns are the vector representations of the (orthogonal) basis states used.

We start off the procedure with a simple Hamiltonian arising from a real-space discretisation of the Schrödinger equation:

$$H_{\mathrm{fixed,fixed}} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}; \quad H_{\mathrm{fixed,free}} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \text{ etc.} \tag{11.44}$$

The information concerning the type of boundary conditions is supplied exclusively in these matrices. They are preserved in the subsequent transformations to larger boxes. Note that we start off with $M = 2$ states. Suppose we envisage keeping $M = 8$ states. The 2×2 matrices will lead to a 4×4 matrices of the large box. Of each of these matrices, we keep the lowest two states, in order to arrive at 8×8 matrices at the next step. From then on we continue glueing these matrices together to 16×16 matrices of which we keep only the lowest two eigenstates, etc. It is important to note that after N steps, the box width is given by $2^{N+1} + 1$. This is because initially, with two boxes, we have a box width of 5, and at each step, the small box width (which starts at 2) is doubled.

Table 11.1. *Energies for the particle in a box after $N = 10$ steps.*

	Energies
E_0	2.3508×10^{-6}
E_1	9.4032×10^{-6}
E_2	2.1157×10^{-5}
E_3	3.7613×10^{-5}

These are the values $n^2\pi^2/L^2$, with $L = 2049$.

PROGRAMMING EXERCISE

You have now enough information for writing a program for this 'real-space quantum renormalisation group'.

Check For the starting matrices given, you should after 10 steps arrive at the spectral values given in Table 11.1.

There is an interesting initiative, called ALPS, to construct a C++ library for quantum algorithms [24, 25]. The programs for this and those in the following sections can be found as examples on the ALPS website.

11.7 The density matrix renormalisation group method

In this section we shall describe how the ideas of the previous section can be extended to many-body systems in one dimension. First of all, we note that the renormalisation procedure of the previous section doubled the box size at each step, and thereby the dimension of the Hilbert space. In the case of a quantum chain, adding a single spin to a spin-1/2 chain already doubles the dimension of the Hilbert space, whereas the dimension of the physical space covered increases only by a small fraction. If we were to double the actual (physical) space, the dimension of the Hilbert space would increase by such a large factor that we would make gigantic steps in complexity, which is bound to fail. Therefore we add only a single site (spin) at a time.

To illustrate how the method works, consider a finite chain of which the lower energy states are properly described by a (small) basis set $|m\rangle$ of size M (that is, m runs from 1 to M). The left end of the chain does not couple to a neighbouring spin: we consider an 'open end' boundary condition there. When we add a new spin, we have states

$$|m, s\rangle \tag{11.45}$$

We have generated a new basis set of size $M \times M_S$ (remember $M_S = 2S + 1$). The main problem now is to find a procedure for reducing this set to a new one of size M. In order to find such a procedure, we must realise ourselves that the system is always *part of a larger system*. The larger system is called the 'universe', the system under consideration is called the 'system', and the remainder (the universe without the system) is called the 'environment'. Universe, system and environment are denoted by U, S and E respectively.

We want the system U to be in the ground state, and the question is how we can represent the state of our system S. The answer lies in the notion of the *density matrix*. Density matrices are described in many quantum textbooks. We shall briefly recall this concept here. The density operator or matrix ρ can be given as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \tag{11.46}$$

The states $|\psi_i\rangle$ are accessible to our system, and they occur with probability p_i. This means that the exact state of the system is not known, but we have a set of 'candidate states' $|\psi_i\rangle$ with probabilities p_i. It is easy to see that for an arbitrary Hermitian operator Q, its expectation value is given as

$$\langle Q\rangle = \mathrm{Tr}(\rho Q). \tag{11.47}$$

We distinguish knowing the quantum state of a system, the *pure state* case, from the situation in which we do not have this knowledge, the *mixed state*. The density matrix corresponding to a pure state is $|\psi\rangle$ is obviously given by

$$\rho = |\psi\rangle\langle\psi|. \tag{11.48}$$

Now consider a system U consisting of two parts, S and E. This system is in a pure state $|\psi^{\mathrm{U}}\rangle$. The basis vectors of the system U can be chosen to be of the form $|\psi_\sigma^{\mathrm{S}}\rangle \otimes |\psi_\epsilon^{\mathrm{E}}\rangle$, where the constituents form complete orthonormal basis sets on S and on E. In the following we shall abbreviate these basis states as $|\sigma\epsilon\rangle$. The behaviour of the part S is completely determined by the expectation values of operators Q^{S} acting within the Hilbert space of S. Now let us evaluate this expectation value for the case where the system U is in the (pure) ground state. This is given by

$$\langle Q^{\mathrm{S}}\rangle = \langle\psi^{\mathrm{U}}|Q^{\mathrm{S}}|\psi^{\mathrm{U}}\rangle. \tag{11.49}$$

Now we expand the ground state in our basis set:

$$|\psi^{\mathrm{U}}\rangle = \sum_{\sigma,\epsilon} C_{\sigma,\epsilon}|\sigma\epsilon\rangle. \tag{11.50}$$

The density matrix then becomes

$$\rho = \sum_{\sigma,\epsilon,\sigma',\epsilon'} C_{\sigma\epsilon}C^*_{\sigma'\epsilon'}|\sigma\epsilon\rangle\langle\sigma'\epsilon'|. \tag{11.51}$$

Now we work out Eq. (11.49) using the basis set $|\sigma\epsilon\rangle$. We obtain

$$\langle Q^{\mathrm{S}}\rangle = \sum_{\sigma,\epsilon,\sigma',\epsilon'} C^*_{\sigma\epsilon} C_{\sigma'\epsilon'} \langle\sigma\epsilon|Q^{\mathrm{S}}|\sigma'\epsilon'\rangle. \tag{11.52}$$

We note that Q^{S} only affects the states of S, so the states of E can immediately be contracted. Using orthonormality of the $|\epsilon\rangle$ we obtain

$$\langle Q^{\mathrm{S}}\rangle = \sum_{\sigma,\epsilon,\sigma'} C^*_{\sigma\epsilon} C_{\sigma'\epsilon} \langle\sigma|Q^{\mathrm{S}}|\sigma'\rangle. \tag{11.53}$$

Noting that

$$\rho^{\mathrm{S}} \equiv \mathrm{Tr}_{\mathrm{E}}\rho = \sum_{\epsilon}\langle\epsilon|\rho|\epsilon\rangle = \sum_{\sigma,\epsilon\sigma'} C_{\sigma\epsilon} C^*_{\sigma'\epsilon} |\sigma\rangle\langle\sigma'|, \tag{11.54}$$

we conclude that

$$\langle Q^{\mathrm{S}}\rangle = \mathrm{Tr}_{\mathrm{S}}(\rho^{\mathrm{S}} Q^{\mathrm{S}}). \tag{11.55}$$

What have we just shown? Starting from a *pure* state of the entire (U) system, we have generated a so-called *reduced density matrix* ρ^{S} for S, which in general describes a mixed state, and which is the most complete description of that system. In other words, when a system S is coupled to an environment E, its state must in general be described as a *mixed* state, although system plus environment together (that is, U) are in a *pure* state.

A system whose state is mixed due to coupling with an environment is said to be *entangled* with that environment. The simplest example of entanglement is a system consisting of two degrees of freedom which can both be in two states, $|0\rangle$ and $|1\rangle$. For a state

$$|\psi\rangle = |00\rangle, \tag{11.56}$$

where the first 0 refers to S and the second one to E, the reduced density matrix is $\rho^{\mathrm{S}} = |0\rangle\langle 0|$ which describes a pure state, reflecting the fact that we know that the system S is in state $|0\rangle$. However, the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{11.57}$$

leads to the reduced density matrix

$$\rho^{\mathrm{S}} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) \tag{11.58}$$

which describes a mixed state. The state $|\psi\rangle$ is therefore an example of an entangled state – this particular state is one of the four *Bell states* [26].

Now let us return to the renormalisation procedure. Knowing that U is in the ground state, we should describe S by a mixed state. But which mixed state? After all, we do not know the ground state of U, and therefore we cannot take the trace

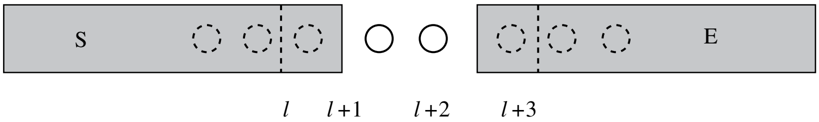

Figure 11.5. Schematic representation of the DMRG procedure. A spin is added to the 'old' system S, which before this addition represented a block of l spins. An environment E is created by reflecting the new S (including the extra spin). Both states are now perfectly known.

over the environment. We can, however, use the information that we have generated on S to create an *artificial* environment E of which all is known. More specifically, we take for E the *reflection* of system S as shown in Figure 11.5. As stated above, we have a basis $|m, s_{l+1}\rangle$ for S. Note that the two indices together belong to S; they can together be viewed as the state $|\sigma\rangle$ of our general discussion of the density matrix above. The reflected system E has a basis $|s_{l+2}, n\rangle$, where we have reversed the indices (and replaced m by n) to represent the structure of E. The combined system U, described by the basis $|m, s_{l+1}, s_{l+2}, n\rangle$ should be in the ground state. If we can find this state, then we can trace out the degrees of freedom of E in order to find the density matrix $\rho^{\rm S}$ describing S. Remember the aim was to find the M most representative basis states of system S, whose Hilbert space has dimension $M \cdot M_S$. We should therefore find the M states which best represent the density matrix $\rho^{\rm S}$. For this we use the idea behind the singular value decomposition, well known from numerical linear algebra [27]: we simply take the M eigenstates of $\rho^{\rm S}$ with the highest eigenvalues (all eigenvalues of the density matrix are positive). In practice, these eigenvalues rapidly decrease so that the errors made in this procedure are very small. The new density matrix is then given as a truncated expansion

$$\rho^{\rm S}(M) = \frac{1}{\sum_{m=1}^{M} \lambda_m} \sum_{m=1}^{M} \lambda_m |m\rangle\langle m| \tag{11.59}$$

where $|m\rangle$ are the eigenstates of $\rho^{\rm S}$. The states $|m\rangle$ are the ones which are carried over to the next stage, analogous to the quantum renormalisation method of the previous section: the Hamiltonian of S is transformed according to

$$H^{\rm S}(M) = V^\dagger H^{\rm S}(M \cdot M_S) V, \tag{11.60}$$

where V contains the M 'most important' eigenstates of $\rho^{\rm S}(M \cdot M_S)$. The fact that this procedure is indeed the best we can follow is addressed in Problems 11.4 and 11.5. The fact that the states kept are derived from an estimate of the density matrix of the system S is reflected in the name *density matrix renormalisation group* (DMRG) for this method.

We still have to address one more question: how can we calculate the Hamiltonian of the system S after a spin has been added to it? After all, we only have some states $|m\rangle$ for which we know the Hamiltonian matrix elements, but we do not know how they relate to individual spins. That would be necessary in order to construct the matrix elements of the Hamiltonian in the basis $|m, s_{l+1}\rangle$, as we need the matrix elements of, for example, the operator $S_{+,l}S_{-,l+1}$, where l is the rightmost spin of the 'old' block S, i.e. before adding the new spin s_{l+1} to it. Note, however, that this matrix element can be written as

$$\langle m, s_{l+1}|S_{+,l}S_{-,l+1}|m', s'_{l+1}\rangle = \langle m|S_{+,l}|m'\rangle\langle s_{l+1}|S_{-,l+1}|s'_{l+1}\rangle, \tag{11.61}$$

a product of two matrix elements. So we must keep track not only of the Hamiltonian in the basis at each step, but also of the rightmost spin operators (in a Heisenberg chain these are the operators S_z, S_+ and S_- at site l). If we want to keep track of physical quantities such as correlation functions, we must keep track of the matrix elements of additional operators in a similar way.

The structure of the algorithm is now as follows:

1. Set up an initial chain of length L, as we did above in the straightforward diagonalisation of the Heisenberg chain. This is the (first approximation to the) system U. The dimension of the Hilbert space is called N.
2. Calculate the ground state of this chain using the Lanczos method.
3. Trace out the degrees of freedom of the right half (E) of the chain U to obtain a reduced density matrix ρ^{S}.
4. Diagonalise the reduced transfer matrix of the left half S of U. Fill a matrix V with the M eigenvectors corresponding to the highest eigenvalues of the density matrix ρ^{S} as columns.
5. Reduce the $\sqrt{N} \times \sqrt{N}$ Hamiltonian H of S to a size $M \times M$ using this matrix V. Do the same for the operators S_l acting on the rightmost spin s_l.
6. Add a spin s_{l+1} to S, so that its Hilbert space now has dimension $\sqrt{N} = M \cdot M_S$.
7. Find the ground state of the $N \times N$ Hamiltonian of the system $\mathrm{U} = \mathrm{S} + \mathrm{E}$ which is spanned by the basis $|m, s_{l+1}, s_{l+2}, n\rangle$.
8. Return to step 3.

If we repeat the algorithm until the density matrix no longer, we have reached a fixed point which corresponds to an infinite chain with open ends.

We now specify how to calculate the product of the Hamiltonian with a n arbitrary vector. At each stage we have the Hamiltonian $H^{\mathrm{S,E}}$ of the system S, E without the spin s_{l+1} added. These Hamiltonians are given in the basis $|m\rangle$. We also have the matrix elements of spin operators $S_{\alpha,l}$ and $S_{\beta,l+3}$ with respect to this basis. The total Hamiltonian is then

$$H^{\mathrm{U}}_{m,s_{l+1},s_{l+2},n;m',s'_{l+1},s'_{l+2},n'} = H^{\mathrm{S}}_{mm'} + \sum_{\alpha\beta} J_{\alpha\beta}[S_{\alpha,l}]_{mm'}[S_{\beta,l+1}]_{s_{l+1},s'_{l+1}}$$
$$+ \sum_{\alpha\beta} J_{\alpha\beta}[S_{\alpha,l}]_{s_{l+1},s'_{l+1}}[S_{\beta,l+1}]_{s_{l+2},s'_{l+2}}$$
$$+ \sum_{\alpha\beta} J_{\alpha\beta}[S_{\alpha,l+2}]_{s_{l+2},s'_{l+2}}[S_{\beta,l+3}]_{n,n'} + H^{\mathrm{E}}_{nn'}, \quad (11.62)$$

where it is understood that every term in this expression must be multiplied by a few Kronecker deltas involving all quantum numbers which do not occur in it. The indices $\alpha\beta$ are zz, $+-$ and $-+$ for the Heisenberg Hamiltonian.

Now we consider the complexity of a matrix–vector multiplication. Evaluating the product of the first term in (11.62) with the vector $|\psi\rangle$ is carried out as follows:

$$\langle m, s_{l+1}, s_{l+2}, n|H^{\mathrm{S}}|\psi\rangle = \sum_{m'} H^{\mathrm{S}}_{mm'}\langle\, m', s_{l+1}, s_{l+2}, n|\psi\rangle, \quad (11.63)$$

and this requires $M^3 M_S^2$ steps, since for each of the $M^2 M_S^2$ elements, a sum over m' must be carried out.

The multiplication for the second term starts by evaluating an intermediate vector $|\phi\rangle$ which is $|\psi\rangle$ multiplied by the term involving $S_{\beta,l+1}$:

$$\langle m, s_{l+1}, s_{l+2}, n|\phi\rangle = \sum_{s'_{l+1}} [S_{\beta,l+1}]_{s_{l+1},s'_{l+1}} \langle m, s'_{l+1}, s_{l+2}, n|\psi\rangle \quad (11.64)$$

which takes $M^2 M_S^3$ steps. Then we multiply the intermediate state $|\phi\rangle$ by the term involving $S_{\alpha,l}$:

$$\langle m, s_{l+1}, s_{l+2}, n|S_{\alpha,l}|\phi\rangle = \sum_{m'} [S_{\alpha,l}]_{m,m'} \langle m', s_{l+1}, s_{l+2}, n|\phi\rangle. \quad (11.65)$$

This is obviously of the same complexity ($M^3 M_S^2$) as the first term.

The remaining terms can now easily be worked out analogous to this procedure.

PROGRAMMING EXERCISE

Write a DMRG program for the one-dimensional Heisenberg chain.

Check You should be able to reproduce the ground state energies for the spin-1/2 and spin-1 chain. The first is analytically known to be $-\ln 2 + 1/2 = -0.443\,147\ldots$ per site, and the second should be found at $-1.40\,148\ldots$ per site. Note that the best way to calculate the energies per site is to subtract the energies found in subsequent steps of the algorithm: this takes into account the energy of the *central* spins and this converges much faster to a fixed value than the total energy divided by the number of sites.

So far, we have considered the simplest DMRG method in fair detail. It is possible to extend the method to periodic chains by not reversing E with respect to S and coupling the right hand side of E to the left hand side of S. Another interesting extension concerns the finite chain instead of the infinite one: sometimes (especially when disorder or incommensurability plays a role) the ground state changes its global structure markedly when varying the chain length, so that studying a fixed length chain is indeed desirable. Another problem in which fixed length chains are very important is finite-size scaling. Although the DMRG method provides a useful way of finding ground states for the infinite chain, finite-size scaling is relevant for finding information about scaling exponents.

The finite-size algorithm works as follows. First the infinite-system algorithm is run for a number of steps, until the universe has the desired length L. Meanwhile, the Hamiltonian and other operators are stored in memory for each system size encountered so far. Then the system S is increased *at the expense of* the environment E, such that U remains constant in size. After E is shrunk to the lowest possible size (one, two or three spins), it is increased again in a stepwise fashion, while reducing the size of S, and this goes on until convergence has been achieved. In more detail, the algorithm for a system of finite size L proceeds along the following steps.

1. Carry out steps of the infinite-size algorithm until the size of the universe (S plus E plus two central spins) is L. During this step, store all the Hamiltonians and other necessary operators (see above) for all sizes in memory.
2. Reduce the matrices (Hamiltonian and necessary spin-operators) of S (which is the left half of the system generated in the previous step) to size M by tracing out E and identifying the most relevant eigenstates of ρ_{S}, followed by a projection of H_{S} and the spin-operators onto the space spanned by these most relevant eigenstates. We now have the Hamiltonian H_{S} for the left half of size $L/2$.
3. Add two spins to H_{S}, and an environment of length $L/2 - 2$ (the Hamiltonian and other matrices of this has been stored in memory). The system S is therefore increased, but E is *decreased*.
4. The procedure of the last two steps is repeated until the system S has size $L - 3$ (that is, E has size 1, but 2 or 3 is also possible). The matrices for S should still be stored for each size.
5. Now *reduce* the size of S in a similar way to the increase of the size of S. The environment therefore grows. For the environment, the stored matrices are used, whereas for the system, the matrices are updated at each step. Carry on until the system S has reached size 1 (and E has size $L - 3$).
6. Repeat the up and down sweeps of the previous steps until convergence has been achieved.

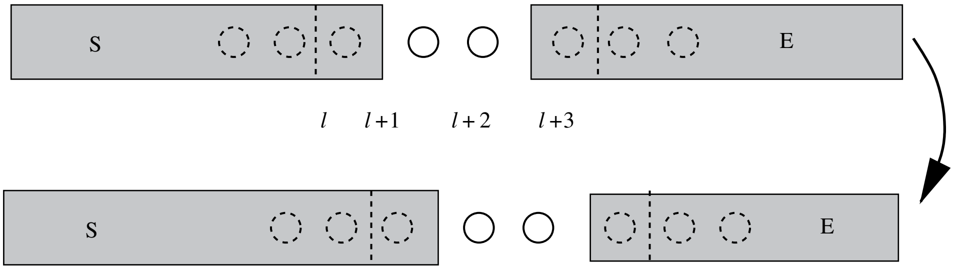

Figure 11.6. The third step in the finite-size DMRG algorithm.

Each time the sweep has reached one of the ends of the chain, the representation of the corresponding end is exact, as it contains only one (or a few) spins. This exact representation is responsible for the improvements in successive sweeps. Step 3 is shown in Figure 11.6. In the particular example of finite-size scaling, the matrices stored for size L can be used for the next size $L-2$. So part of the work is used later again. Interestingly, the dynamics of polymer chains, which is governed by a Master equation, can also be treated within the DMRG approach, and for reptation and electrophoresis, useful results have been obtained [28, 29].

The major effort in constructing a program for the finite-size chain consists of storing the relevant operator matrices H, S_z etc. for the various sizes in memory. The algorithm can then be implemented straightforwardly. As a check, you can verify whether the ground state energy found corresponds to that found for finite Heisenberg chains in Section 11.5.1.

A major point is whether the DMRG method is applicable to systems in more than one dimension. This turns out to be difficult. Trials have been made in two directions. The first is to consider a two-dimensional system as a one-dimensional chain, 'wrapped up' to fill the plane [30]. The second is to formulate the Hubbard Hamiltonian in k-space, which renders the hopping term diagonal and the Coulomb interaction no longer local. Reviews concerning DMRG can be found in Refs. [31–33].

Exercises

11.1 The Pauli matrix σ_x is given as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Using $\sigma_x^2 = 1$, show that

$$\mathrm{e}^{\beta L \sigma_x} = \cosh(\beta L) + \sigma_x \sinh(\beta L).$$

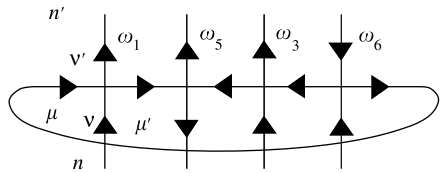

Figure 11.7. Action of the transfer matrix in the six-vertex model. The transfer matrix connects two adjacent horizontal rows of vertical arrows – the possible configurations of horizontal arrows are summed over. The index μ indicates the orientation of the leftmost arrow, and μ' represents the direction of the horizontal arrows in the iterative multiplication process. In this figure, it labels the second horizontal link, as in the first step of the multiplication. Only a single configuration of vertices is shown.

Comparing this with the transfer matrix of the one-dimensional Ising model with zero magnetic field, show that

$$\tanh(\beta L) = \exp(-2\beta J).$$

11.2 In this problem we consider the implementation of the transfer matrix method for the six-vertex model. The vertices and weights are represented in Figure 11.3. The transfer matrix connects a row of vertical arrows to the next one as in Figure 11.7. This implies that for each pair of adjacent rows of vertical arrows a sum must be performed over all possible configurations of horizontal arrows that are compatible with the vertical ones, in the sense that the number of ingoing arrows equals the number of outgoing ones at each site.

The multiplication of the transfer matrix with an arbitrary vector ϕ is again carried out site by site, similar to the procedure followed for the Ising model. Let us first consider the multiplication including the first (leftmost) vertex ($j = 1$). It turns out to be necessary to introduce a vector $\psi(n, \mu, \mu')$ in the multiplication routine. Here, n represents a row configuration of vertical arrows and μ and μ' are indices assuming the values 0 and 1 – they represent the direction of the leftmost horizontal arrow and the horizontal arrow on the link connecting sites j and $j + 1$ ($\mu = 0$ denotes a right-pointing arrow, $\mu = 1$ a left-pointing one), which is the second horizontal arrow from the left as we are considering the first (leftmost) vertex.

As the two horizontal arrows of the first site are known, there is a unique relation between the upper and lower vertical arrows. These arrows correspond to the least significant bits ν and ν' which are given by n mod 2 and n' mod 2 respectively ($\nu = 0$ is an upward-pointing arrow, $\nu = 1$ a downward-pointing one). Therefore we have

$$\nu + \mu = \nu' + \mu'. \tag{E11.1}$$

This leads to the following code for the multiplication routine involving the first (leftmost) vertex:

```
FOR n = 0 TO 2^(M-1) - 1 DO {n runs over half the number of
                             row states }
l = 2n;                     {l is a number with least significant
                             bit equal to 0}
   ψ'(l,0,0) = ω1 ψ(n);
   ψ'(l,1,1) = ω3 ψ(n);
   ψ'(l+1,1,0) = ω6 ψ(n);
   l = l + 1                {l is now a number with least
                             significant bit equal to 1}
   ψ'(l,0,0) = ω2 ψ(n);
   ψ'(l,1,1) = ω4 ψ(n);
   ψ'(l-1,0,1) = ω6 ψ(n);
END FOR
```

Now the second vertex must be treated. We start from the vector $\psi'(n,\mu,\mu')$ and calculate the next vector $\psi''(n,\mu,\mu')$. The index μ remains the leftmost horizontal arrow (we need this to ensure PBC in the end), and the second corresponds to the link connecting sites 1 and 2. The procedure is analogous to that for the first vertex.

After completing the sequence through the full row, we calculate the resulting vector $\phi(n)$ from the last vector $\psi^{\text{last}}(n,\mu,\mu')$ obtained in the foregoing procedure by taking the trace:

```
FOR N = 0 TO 2^(M-1)
   φ(n) = ψ^last(n,0,0) + ψ^last(n,1,1);
END FOR
```

The results can be analysed in a similar fashion to the Ising model. For the weights of the F-model, the central charge should be equal to 1.

11.3 The Hamiltonian of the periodic Heisenberg chain can be written as

$$H = J\sum_{l=1}^{L} \mathbf{S}_l \mathbf{S}_{l+1}$$

with $\mathbf{S}_{L+1} \equiv \mathbf{S}_1$.

(a) Show that this can be transformed into

$$H = J\sum_{l=1}^{L}\left[\frac{1}{2}(S_{+,l}S_{-,l+1} + S_{-,l}S_{+,l+1}) + S_{z,l}S_{z,l+1}\right],$$

where $S_\pm = S_x \pm iS_y$. Give the matrix forms of $S_\pm$ for $s = 1/2$ and $s = 1$.

(b) Show that $S_z = \sum_{l=1}^{L} S_{z,l}$ commutes with H. Also show that

$$S^2 = \left(\sum_{l=1}^{L} \mathbf{S}_l \right)^2$$

commutes with H.

11.4 In the DMRG, we take the set of eigenvectors $|m\rangle$ with the highest eigenvalues of ρ^{S} as representatives of the system S. To show that this is indeed the best procedure, we consider the representation of an arbitrary state $|\psi\rangle$, of the universe U, which can be expanded as

$$|\psi\rangle = \sum_{mn} C_{mn} |mn\rangle,$$

where, as usual, m labels a basis vector of S and n one of E.

We now replace the basis $|m\rangle$ of S by a smaller basis $|u_\alpha\rangle$, $\alpha = 1, \ldots, M$ where we choose the orthonormal basis vectors $|u_\alpha\rangle$ such that the state $|\psi\rangle$ can be optimally represented by them. This means that the norm of the difference between the exact and the expanded state

$$\left| \sum_{mn} C_{mn} |mn\rangle - \sum_{\alpha n} D_{\alpha n} |u_\alpha n\rangle \right|^2$$

is minimal.

We can expand the basis vectors $|u_\alpha\rangle$ in terms of the basis $|m\rangle$:

$$|u_\alpha\rangle = \sum_m u_{\alpha m} |m\rangle.$$

(a) Formulate the error (i.e. the norm of the difference) in terms of the coefficients $D_{\alpha n}$ and $u_{\alpha m}$. Minimise this error with respect to both the $D_{\alpha n}$ and the $u_{\alpha m}$ and show that this leads to the equation

$$-\sum_m C_{nm} u_{\alpha m} + \sum_{m\alpha'} D_{\alpha n} u_{\alpha m} u_{\alpha' m} = 0$$

and

$$-\sum_n C_{mn} D_{\alpha n} + \sum_{n\alpha'} D_{\alpha n} D_{\alpha' n} u_{\alpha' m} = 0.$$

(b) Use the orthonormality of the $|u_\alpha\rangle$ to infer from the first equation:

$$\sum_m C_{mn} u_{\alpha m} = D_{\alpha n}.$$

(c) Substitute this into the second equation to find

$$\sum_{nm'} C_{mn} C_{m'n} u_{\alpha m'} = \sum_{nm'm''} C_{m'n} C_{m''n} u_{\alpha m'} u_{\alpha' m''} u_{\alpha' m}.$$

Using

$$\rho^{\mathrm{S}}_{mm'} = \sum_n C_{mn} C_{m'n},$$